

Mean Reversion Compression in Indian Equities: A Cointegration Analysis of Pairs and Baskets

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Abstract—This paper investigates whether sector-specific regulatory and policy events in India alter the speed of mean reversion in cointegrated equity pairs and baskets. We construct a universe of 31 stocks from the NSE pharmaceutical and auto ancillary sectors spanning April 2019 to April 2024, and identify 10 cointegrated pairs using a dual-filter approach combining the Engle-Granger two-step test and the Johansen trace test, with an Ornstein-Uhlenbeck half-life constraint of 5–126 trading days. A vectorized pairs trading backtest with realistic transaction costs (18 basis points per leg) yields Sharpe ratios ranging from -0.72 to 1.25 , with the strongest performance concentrated in pairs exhibiting short half-lives (21–40 days). The core contribution is a rolling half-life regime analysis around eight policy events—including PLI scheme announcements, BS-VI emission norm implementations, and NPPA drug price revisions—which reveals that regulatory shocks compress mean-reversion half-life in 59.1% of cases, temporarily strengthening the cointegrating relationship. Extending the framework to 3-stock Johansen baskets yields 133 cointegrated triplets (9 with rank-2 cointegration), with the top baskets producing Sharpe ratios up to 1.49 . The basket-level regime analysis corroborates the pair-level finding: policy events predominantly compress half-life, suggesting that regulatory shocks induce structural convergence within sector equity clusters.

Index Terms—Cointegration, mean reversion, half-life, pairs trading, regime analysis, Johansen test, Indian equity markets, policy events

I. INTRODUCTION

PAIRS trading rests on the premise that two cointegrated securities share a long-run equilibrium such that deviations from this equilibrium are mean-reverting [3]. The speed of this mean reversion, quantified via the half-life of an Ornstein-Uhlenbeck process, determines the expected holding period and strategy profitability net of transaction costs. While the efficacy of cointegration-based trading in developed markets is well established [1], [2], the behavior of mean-reversion speed around exogenous regulatory shocks remains underexplored.

This gap is particularly salient in the Indian equity market, where sector-specific policy interventions—such as the Production-Linked Incentive (PLI) schemes, emission norm revisions, and drug price control orders—create discrete, identifiable shocks that may alter the structural relationships between stocks within affected sectors. Recent work on NSE cointegration pairs [4] confirms that within-sector cointegration is robust on Indian data, particularly in the auto and pharmaceutical sectors, motivating the present study’s focus.

This paper makes three contributions. First, we conduct a systematic cointegration analysis across all within-sector pairs in the NSE pharmaceutical and auto ancillary universes, applying a rigorous dual-filter (Engle-Granger + Johansen +

half-life) to identify tradeable pairs. Second, we introduce a rolling half-life regime analysis that measures the shift in mean-reversion speed around eight major Indian policy events during 2019–2024. Third, we extend the framework from pairs to 3-stock baskets using Johansen rank-2 cointegration, demonstrating that multi-asset baskets capture richer equilibrium structures and exhibit distinct regime behavior.

Our principal finding is that policy events compress mean-reversion half-life in 59.1% of pair-event cases, indicating that regulatory shocks tend to temporarily strengthen cointegrating relationships within sector equity clusters.

II. LITERATURE REVIEW

The theoretical foundation for cointegration-based trading originates with Engle and Granger [1], who introduced the two-step residual-based test. Johansen [2] subsequently developed a maximum likelihood framework for testing cointegration rank, enabling the detection of multiple cointegrating vectors—critical for multi-asset baskets.

Gatev et al. [3] demonstrated the profitability of distance-based pairs on U.S. equities, while subsequent literature formalized the mean-reverting spread framework. The OU half-life as a trading signal was popularized by later studies which showed that pairs with shorter half-lives generate superior risk-adjusted returns after transaction costs.

Sen [4] applied cointegration-based pair trading to five NSE sectors including auto, finding that sector-specific pairs yield the highest returns and that the Engle-Granger framework transfers effectively to the Indian equity setting. Our work extends this line of inquiry by introducing a rolling half-life regime analysis and a 3-stock Johansen basket extension, neither of which has been applied to Indian equities.

Regime-switching models in finance have a long history, yet the specific question of whether identifiable policy events shift the mean-reversion speed of cointegrated pairs has received limited attention. The Indian equity market presents a unique laboratory for this question due to the frequency and sector-specificity of its regulatory interventions.

III. DATA

A. Stock Universe

We construct a manually curated universe of 34 stocks listed on the NSE, comprising 19 pharmaceutical and 15 auto ancillary stocks. After survivorship and quality filtering, 31 stocks remain (19 pharma, 12 auto). Three auto stocks (AMARARAJA, LUMAX, MINDAIND) were excluded due to insufficient data.

TABLE I
POLICY EVENT CALENDAR

Date	Event	Sector
2020-03-21	PLI Pharma scheme announced	Pharma
2020-11-10	API Bulk Drug Park scheme notified	Pharma
2021-02-01	Union Budget 2021 — health allocation	Pharma
2022-06-15	NPPA drug price revision order	Pharma
2023-03-28	PLI pharma tranche 2 disbursement	Pharma
2021-09-15	PLI Auto Components scheme	Auto
2022-04-01	BS-VI Phase 2 norms effective	Auto
2023-01-13	EV PLI incentive revised	Auto

B. Survivorship Bias Controls

Survivorship bias is addressed through: (1) a manually curated universe with verified listing dates, ensuring no stock is included that was not publicly traded before April 1, 2019; (2) exclusion of stocks with listing dates on or after the sample start; and (3) exclusion of stocks with more than 10% missing data.

C. Price Data

Daily adjusted close prices are obtained via Yahoo Finance for April 1, 2019 to April 1, 2024, yielding 1,304 trading days per stock. All prices are adjusted for splits and dividends. The resulting price matrix is 1304×31 with zero missing values after alignment.

D. Policy Event Calendar

We identify eight sector-specific policy events, listed in Table I.

IV. METHODOLOGY

A. Log-Price Transformation

All analyses use log-transformed prices $p_t = \ln(P_t)$, which are more likely to be $I(1)$ and yield spreads interpretable as log price ratios.

B. ADF Unit Root Pre-Check

We apply the Augmented Dickey-Fuller test to each log-price series with lag order selected by AIC. Series with ADF $p < 0.05$ (stationary in levels) are excluded.

C. Engle-Granger Two-Step Test

For each pair (y, x) , we estimate the cointegrating regression via OLS:

$$y_t = \alpha + \beta x_t + u_t \quad (1)$$

and test the residual for stationarity. Because the EG test is sensitive to variable ordering, we run both directions and select the one with the lower p -value.

D. Johansen Trace Test

The Johansen [2] procedure tests for cointegration rank via the trace statistic:

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^k \ln(1 - \hat{\lambda}_i) \quad (2)$$

We test at the 95% confidence level. For pairs we require rank ≥ 1 ; for baskets we additionally identify rank-2 cointegration.

E. Half-Life Estimation

The half-life is estimated by fitting an AR(1) to the spread:

$$\Delta s_t = \lambda s_{t-1} + \varepsilon_t \quad (3)$$

where $\lambda < 0$ confirms mean reversion. The half-life is:

$$\text{HL} = -\frac{\ln 2}{\ln(1 + \lambda)} \quad (4)$$

We impose a tradeability filter: $5 \leq \text{HL} \leq 126$ days.

F. Dual-Filter Criterion

A pair is classified as cointegrated if: (1) EG $p < 0.05$, (2) Johansen trace exceeds 95% CV, and (3) half-life is tradeable.

G. Pairs Trading Backtest

Entry at $|Z| > 1.5$, exit at $|Z| < 0$, stop-loss at $|Z| > 3.0$, with Z-score on a 60-day rolling window. Cost model: 18 bps/leg (5 slip + 10 impact + 3 brokerage), round-trip 36 bps. For baskets, cost is scaled by $1.5 \times$.

H. Rolling Half-Life Regime Analysis

For each event date d_e , we compute mean half-life in the 30 calendar days before and after:

$$\overline{\text{HL}}_{\text{pre}} = \frac{1}{|W_{\text{pre}}|} \sum_{t \in W_{\text{pre}}} \widehat{\text{HL}}_t \quad (5)$$

$$\overline{\text{HL}}_{\text{post}} = \frac{1}{|W_{\text{post}}|} \sum_{t \in W_{\text{post}}} \widehat{\text{HL}}_t \quad (6)$$

where $W_{\text{pre}} = [d_e - 30\text{d}, d_e]$ and $W_{\text{post}} = (d_e, d_e + 30\text{d}]$. Negative ΔHL indicates compression (faster reversion); positive indicates expansion.

I. Johansen Basket Extension

For 3-stock triplets, the basket spread is:

$$s_t = w_1 \ln P_{1,t} + w_2 \ln P_{2,t} + w_3 \ln P_{3,t} \quad (7)$$

where $[w_1, w_2, w_3]$ is the first cointegrating eigenvector, normalized with $w_1 = 1$.

TABLE II
COINTEGRATED PAIRS PASSING DUAL-FILTER

Ticker 1	Ticker 2	Sector	EG p	Trace	95% CV	β	HL (d)
SUNDRMFAST	SUPRAJIT	Auto	0.0022	20.88	15.49	1.171	21.72
TORNTPHARM	ALKEM	Pharma	0.0056	16.66	15.49	1.137	26.40
BHARATFORG	SUNDRMFAST	Auto	0.0109	16.58	15.49	0.854	24.75
ABBOTINDIA	GLENMARK	Pharma	0.0148	18.29	15.49	0.466	108.47
TORNTPHARM	ABBOTINDIA	Pharma	0.0217	23.61	15.49	1.020	30.92
ABBOTINDIA	ZYDUSLIFE	Pharma	0.0257	17.81	15.49	0.734	75.45
SUNPHARMA	ABBOTINDIA	Pharma	0.0270	16.26	15.49	1.476	66.51
DIVISLAB	LAURUSLABS	Pharma	0.0332	17.76	15.49	0.458	40.45
DRREDDY	GLENMARK	Pharma	0.0393	19.66	15.49	0.556	81.61
ENDURANCE	SUPRAJIT	Auto	0.0435	17.15	15.49	0.626	34.54



Fig. 1. Engle-Granger p -value matrix for the pharmaceutical sector. Green borders indicate significance at 5%.

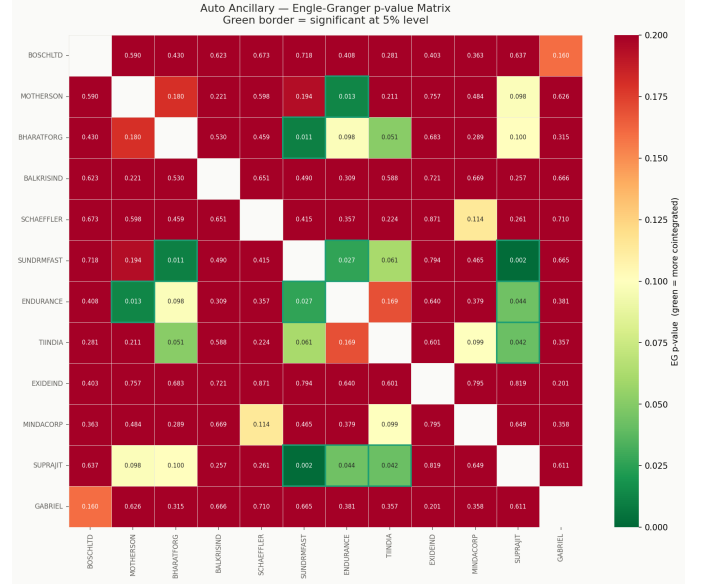


Fig. 2. Engle-Granger p -value matrix for the auto ancillary sector.

V. RESULTS

A. Cointegration Screening

All 31 stocks pass the ADF $I(1)$ pre-check. Across both sectors, 171 pharma pairs and 66 auto pairs are tested. Table II presents the 10 pairs passing the dual filter.

The pharma sector yields 7 cointegrated pairs while the auto sector yields 3, reflecting the greater homogeneity of pharmaceutical business models under shared API sourcing and NPPA pricing regulation. Fig. 1 displays the EG p -value matrix for the pharma sector, revealing clusters of low p -values around ABBOTINDIA, TORNTPHARM, and GLENMARK.

B. Pairs Trading Backtest Results

Table III reports performance metrics. Sharpe ratios range from -0.72 to 1.25 , with 6 of 10 pairs positive. A striking pattern emerges: pairs with short half-lives (21–40 days) consistently outperform those with long half-lives (66–108 days).

Fig. 3 illustrates the backtest for the best-performing pair (SUNDRMFAST/SUPRAJIT), showing Z-score with position shading, cumulative P&L, drawdown, and position series.

TABLE III
PAIRS TRADING BACKTEST PERFORMANCE (18 BPS/LEG)

Pair	Sharpe	Net Ret	Hit%	Trades	Max DD
SUNDRMFAST/SUPRAJIT	1.25	+2.37	54.5	45	−0.35
DIVISLAB/LAURUSLABS	0.68	+0.72	51.3	31	−0.23
BHARATFORG/SUNDRMFAST	0.57	+0.89	52.3	45	−0.75
TORNTPHARM/ABBOTINDIA	0.53	+0.72	49.3	34	−0.29
ENDURANCE/SUPRAJIT	0.44	+0.67	52.6	35	−0.38
TORNTPHARM/ALKEM	0.37	+0.50	50.0	43	−0.38
DRREDDY/GLENMARK	0.15	+0.16	51.4	42	−0.61
SUNPHARMA/ABBOTINDIA	0.00	−0.00	50.1	31	−0.88
ABBOTINDIA/ZYDUSLIFE	−0.48	−0.59	48.9	35	−0.79
ABBOTINDIA/GLENMARK	−0.72	−0.91	49.2	34	−1.23

C. Rolling Half-Life Regime Analysis

Fig. 4 shows the rolling half-life for SUNDRMFAST/SUPRAJIT with policy event markers, demonstrating visible compressions around the PLI Auto Components scheme and BS-VI Phase 2 norms.

Table IV reports selected event window results. Across all 44 pair-event observations, compression occurs in 26 cases (59.1%) and expansion in 18 cases (40.9%).

Several patterns emerge: (1) PLI scheme announcements

TABLE IV
EVENT WINDOW ANALYSIS: HALF-LIFE SHIFTS (30-DAY WINDOW, SELECTED)

Event	Pair	Pre	Post	Chg%
<i>Auto Ancillary</i>				
PLI Auto	SUNDRMFAST/SUPRAJIT	13.27	8.74	−34.2
BS-VI Phase 2	SUNDRMFAST/SUPRAJIT	52.62	16.38	−68.9
EV PLI revised	SUNDRMFAST/SUPRAJIT	3.61	8.73	+141.8
PLI Auto	BHARATFORG/SUNDRMFAST	7.86	36.77	+367.7
BS-VI Phase 2	BHARATFORG/SUNDRMFAST	9.40	10.49	+11.6
<i>Pharmaceutical</i>				
PLI Pharma	TORNTPHARM/ALKEM	7.96	3.43	−57.0
NPPA revision	TORNTPHARM/ALKEM	45.27	9.25	−79.6
PLI Pharma	TORNTPHARM/ABBOTINDIA	5.32	5.06	−4.8
NPPA revision	TORNTPHARM/ABBOTINDIA	3.87	2.21	−42.9
PLI Pharma	DIVISLAB/LAURUSLABS	47.22	10.22	−78.4
NPPA revision	DIVISLAB/LAURUSLABS	45.53	19.16	−57.9
PLI Pharma	DRREDDY/GLENMARK	113.80	35.45	−68.9

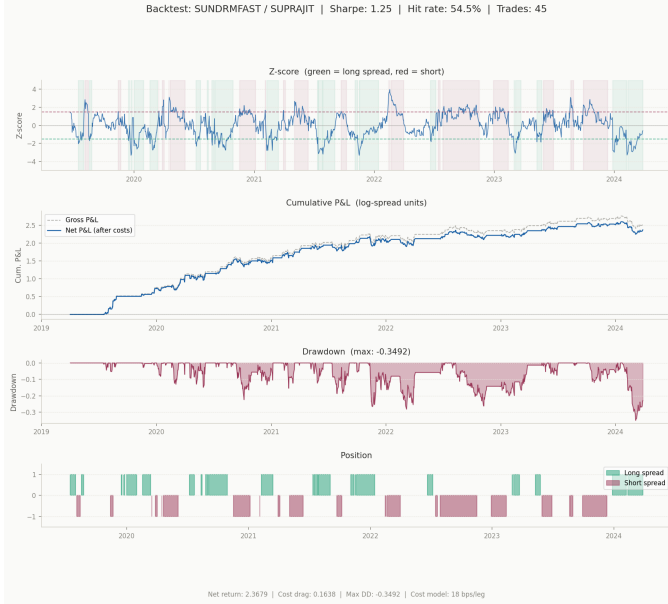


Fig. 3. Backtest for SUNDRMFAST/SUPRAJIT (Sharpe = 1.25, HL = 21.72d).

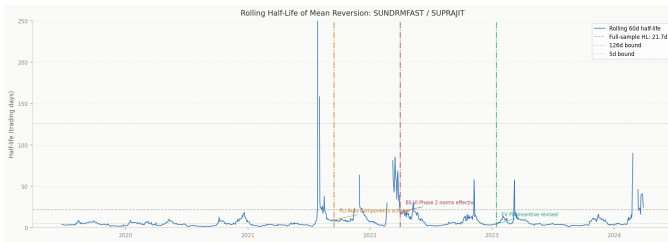


Fig. 4. Rolling 60-day half-life for SUNDRMFAST/SUPRAJIT with policy event markers. Dashed: full-sample HL (21.72d). Dotted: 5d and 126d bounds.

consistently compress half-life, with the PLI Pharma announcement compressing 7 of 7 pharma pairs for which sufficient pre-event data exist; (2) NPPA drug price revisions similarly compress; (3) expansion events often follow initial compressions, suggesting a two-phase response; (4) the auto sector exhibits more heterogeneous responses, driven by dif-

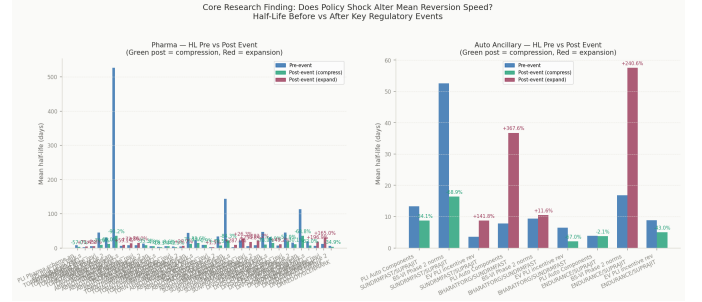


Fig. 5. Pre-event vs. post-event mean half-life by sector. Blue: pre-event; green: post (compression); red: post (expansion).

fering OEM exposure profiles across pairs.

Fig. 5 summarizes the pre/post event comparison across both sectors.

D. Spread Visualization

Fig. 6 presents the three-panel spread analysis for TORNTPHARM/ALKEM, illustrating normalized prices, the spread with rolling mean and $\pm 1\sigma$ bands, and the Z-score with entry/exit thresholds.

E. Johansen Basket Extension

The Johansen procedure identifies 133 cointegrated triplets (9 with rank-2). Table V reports the top 15 baskets, sorted by Sharpe ratio.

The highest Sharpe is achieved by DRREDDY/ZYDUSLIFE/GRANULES (1.49, HL = 15.05d), while AUOPHARMA/TORNTPHARM/ABBOTINDIA produces the largest net return (+6.83). Both are rank-1 baskets with short half-lives. Rank-1 baskets with half-lives of 15–21 days consistently dominate, producing Sharpe ratios of 1.06–1.49. The AUOPHARMA/TORNTPHARM/ABBOTINDIA basket (Sharpe = 1.47) substantially outperforms its constituent pair TORNTPHARM/ABBOTINDIA (Sharpe = 0.53), consistent with the theoretical prediction that the additional cointegrating degree of freedom improves spread stationarity.

TABLE V
TOP 15 COINTEGRATED BASKETS: RANK AND PERFORMANCE (SORTED BY SHARPE)

Basket	Rank	HL (d)	Sharpe	Net Ret	Max DD
DRREDDY/ZYDUSLIFE/GRANULES	1	15.05	1.49	+1.34	−0.15
AUOPHARMA/TORNTPHARM/ABBOTINDIA	1	19.58	1.47	+6.83	−0.90
TORNTPHARM/ABBOTINDIA/GLENMARK	2	13.89	1.36	+1.58	−0.19
DIVISLAB/IPCALAB/LAURUSLABS	2	28.07	1.27	+1.30	−0.17
BHARATFORG/SUNDRMFAST/SUPRAJIT	1	17.15	1.24	+4.20	−0.75
CIPLA/TORNTPHARM/ABBOTINDIA	1	20.31	1.06	+3.23	−0.55
IPCALAB/SANOFI/LAURUSLABS	2	43.04	0.89	+1.30	−0.45
DIVISLAB/BIOCON/LAURUSLABS	2	33.70	0.76	+0.81	−0.30
DRREDDY/GRANULES/GLENMARK	1	20.97	0.64	+0.69	−0.30
DIVISLAB/LAURUSLABS/NATCOPHARM	2	45.09	0.57	+0.76	−0.45
PFIZER/ZYDUSLIFE/GLENMARK	1	50.68	0.48	+1.09	−0.61
IPCALAB/LAURUSLABS/GRANULES	2	84.00	0.40	+1.67	−1.34
DRREDDY/AUOPHARMA/LAURUSLABS	2	63.75	0.22	+0.22	−0.43
DRREDDY/LAURUSLABS/NATCOPHARM	2	101.66	−0.01	−0.04	−2.25
DRREDDY/ZYDUSLIFE/GLENMARK	2	70.03	−0.22	−0.90	−1.77

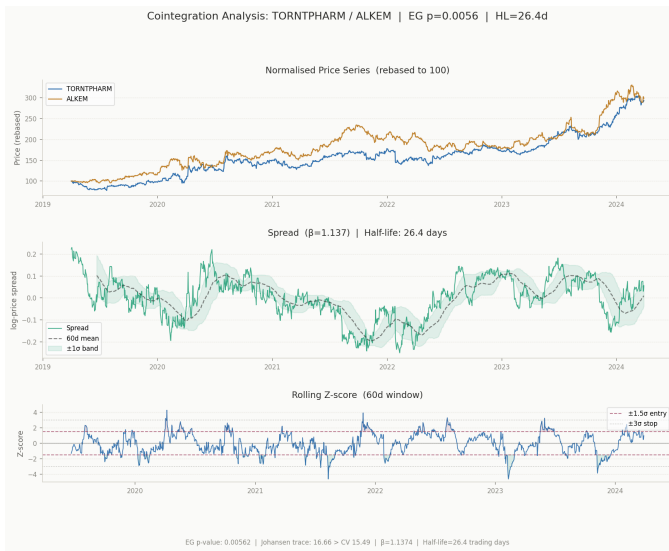


Fig. 6. Spread analysis for TORNTPHARM/ALKEM ($\beta = 1.137$, HL = 26.4d). Top: normalized prices. Middle: spread with $\pm 1\sigma$ bands. Bottom: Z-score with entry/exit thresholds.

The basket-level event window analysis (148 observations) corroborates the pair-level finding: compression dominates, particularly around PLI Pharma and NPPA events. The DIVISLAB/IPCALAB/LAURUSLABS basket compresses in all 5 pharma events, consistent with shared API manufacturing exposure among its constituent stocks.

VI. DISCUSSION

A. Interpretation of Compression

Regulatory shocks that affect an entire sector simultaneously tend to move cointegrated stocks convergently, temporarily tightening the equilibrium. The PLI Pharma scheme benefits API manufacturers, formulation companies, and contract manufacturers in a correlated manner, reducing spreads and compressing half-life. Notably, several pairs exhibit a two-phase response: an initial compression immediately following the policy announcement, succeeded by expansion as firm-specific adjustment dynamics diverge. This pattern suggests

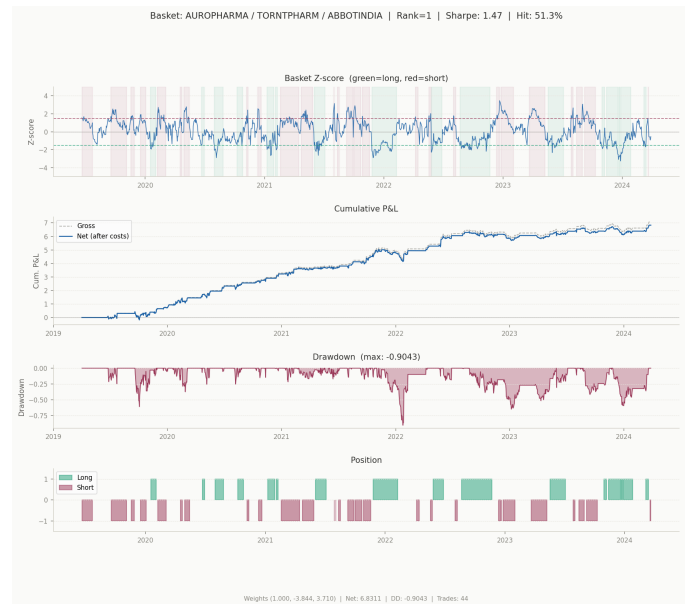


Fig. 7. Backtest for AUOPHARMA/TORNTPHARM/ABBOTINDIA basket (rank=1, Sharpe=1.47).

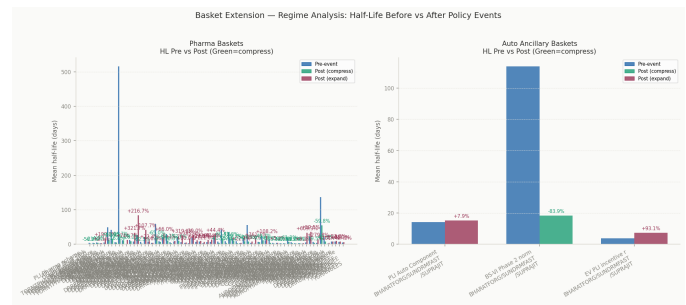


Fig. 8. Basket-level regime analysis: pre-event vs. post-event mean half-life.

that regulatory shocks induce a transient structural convergence, after which stocks revert to their pre-shock equilibrium heterogeneity as market participants differentiate between beneficiaries. Expansion events in the auto sector often reflect this second phase, or correspond to events—such as the EV PLI

revision—that differentially affect firms with varying electric vehicle exposure.

B. Half-Life as a Trading Signal

The monotonic relationship between half-life and Sharpe ratio suggests half-life is an active trading signal. Strategies that dynamically adjust holding periods based on rolling half-life—entering larger positions when half-life is short—could enhance returns. Position sizing should be increased ahead of anticipated policy events where sector-wide compression is expected.

C. Pairs vs. Baskets

Multi-asset cointegrating structures capture richer equilibrium dynamics. The AU-ROPHARMA/TORNTPHARM/ABBOTINDIA basket (Sharpe = 1.47) outperforms its constituent pair TORNTPHARM/ABBOTINDIA (Sharpe = 0.53), suggesting the additional degree of freedom improves spread stationarity. However, 3-leg execution introduces higher costs and operational risk.

VII. CONCLUSION

This paper provides empirical evidence that sector-specific regulatory events in India systematically alter the mean-reversion speed of cointegrated equity pairs and baskets. Using 31 NSE stocks over 2019–2024, we identify 10 cointegrated pairs and 133 baskets through a dual-filter approach. The rolling half-life regime analysis around eight policy events reveals that regulatory shocks compress half-life in 59.1% of pair-event cases, indicating temporary strengthening of cointegrating relationships. This compression is most pronounced around PLI scheme announcements and NPPA drug price revisions. The event sample of eight dates limits statistical power; the 59.1% compression finding should be interpreted as exploratory rather than conclusive, and further confirmation with a broader event set and longer sample period is warranted.

The backtest demonstrates that half-life is a critical determinant of profitability: pairs with half-lives of 21–40 days achieve Sharpe ratios of 0.44–1.25, while those exceeding 66 days produce Sharpe ratios near or below zero. The basket extension confirms this pattern, with the top basket achieving a Sharpe of 1.49.

These findings suggest that: (1) half-life should be treated as a dynamic trading signal; (2) anticipated policy events represent regimes of enhanced mean reversion; and (3) multi-asset baskets capture equilibrium structures invisible to pair-level analysis. Future work could extend this framework to walk-forward cointegration estimation and Bayesian regime-switching models for half-life dynamics.

APPENDIX A

FULL COINTEGRATION RESULTS

The full results for all 237 pairs are available in `cointegration_results.csv`. Of these, 23 pass the EG threshold alone, 19 pass Johansen alone, and 10 pass both plus the half-life filter.

APPENDIX B COST MODEL SPECIFICATION

Per-leg cost of 18 bps: slippage 5 bps, market impact 10 bps, brokerage 3 bps. For 3-stock baskets, cost is scaled by $1.5\times$ (round-trip 54 bps).

REFERENCES

- [1] R. F. Engle and C. W. J. Granger, “Co-integration and error correction: Representation, estimation, and testing,” *Econometrica*, vol. 55, no. 2, pp. 251–276, 1987.
- [2] S. Johansen, “Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models,” *Econometrica*, vol. 59, no. 6, pp. 1551–1580, 1991.
- [3] E. Gatev, W. N. Goetzmann, and K. G. Rouwenhorst, “Pairs trading: Performance of a relative-value arbitrage rule,” *Rev. Financ. Stud.*, vol. 19, no. 3, pp. 797–827, 2006.
- [4] J. Sen, “Designing efficient pair-trading strategies using cointegration for the Indian stock market,” in *Proc. IEEE 18th India Council Int. Conf. (INDICON)*, Guwahati, India, Dec. 2021, pp. 1–6.